Arrays in Js

Arrays

- In JavaScript, array is a single variable that is used to store different elements.
- It is often used when we want to store list of elements and access them by a single variable.
- Unlike most languages where array is a reference to the multiple variable, in JavaScript array is a single variable that stores multiple elements

    Eg: const fruits=[“Banana”, “Apple”, ”Grapes”]

    const a=[05,”Anju”, true]

Declaration of an Array

There are basically two ways to declare an array.

Example:

```
var vehicle = [ ]; // method 1
var House = new Array(); // method 2
```

Initialization of an Array

Example (for Method 1):

// Initializing while declaring

```
var vehicle = ["car", "bus", "bike", "auto"];
```

Example (for Method 2):

// Initializing while declaring

// Creates an array having elements 10, 20, 30, 40, 50

```
var vehicle = new Array(10, 20, 30, 40, 50);
```

Creates an array of 5 undefined elements

```
var vehicle1 = new Array(5);

// Now assign values

vehicle[0]="car"

vehicle[1]="bus"

vehicle[2]="bike"

vehicle[3]="auto"

vehicle[4]="cycle"
```

Creates an array with element car

```
var vehicle = new Array("car");
```

An array in JavaScript can hold different elements

We can store Numbers, Strings and Boolean in a single array.

Example:

```
// Storing number, boolean, strings in an Array

var vehicle = ["car", 2500000, "bike", 500000, "sale", true];
```

Properties of an Array

- **Length property of an Array**

```
var len = vehicle.length;
```

- **Fetch every element of the array**

```
console.log(vehicle);
```

- **Fetch one by one elements of the array**

```
for(let i of vehicle) {
    console.log(i);
}
```

Properties of an Array

To insert a new element in an array

-
```
console.log(vehicle.push("auto"));

console.log(vehicle);
```

of keyword - to get values in an array

-
```
for(let i of vehicle){

console.log(i);}
```

in keyword - to get index position of values stored in an array

-
```
for(let index in vehicle){

console.log(index);

}
```

- **Normal for loop for displaying values in an array**

```
for(i=0; i<vehicle.length; i++){
    console.log(vehicle[i]);
}
```

- **Changing Position of an element**

```
let numbers=[7,2,40,9]
numbers[2]=8
```

Tasks

1. Write a note on array methods in javascript with example. **Note:You should find all the methods and try to implement them.**
2. Create an array of numbers and add numbers to this array and Keep adding numbers to the array in the even number format
3. filter the numbers divisible by 10 from a given array Eg: [10,25,67,45,70,36,50]
4. Create an array of square of given number eg: [3,4,5,6,7]
5. Write a function to remove all duplicate elements from an array.
6. Write a function to reverse an array in-place (without creating a new array).
7. Write a function to find the intersection of two arrays
8. Write a function to remove all falsy values (e.g., null, undefined, 0, false) from an array.
9. Write a function to shuffle the elements of an array randomly.
10. Write a function to find the difference between two arrays (elements in one array but not in the other).